| PRODUCT | Memory | Notes |
|---|---|---|
| AD1460 | 4 Mbit ×4 | Quad-SHARC, Four ADSP-21060's in the same module; provides an incredible 480 MFLOPS in only 2.05"×2.05"×0.16". |
| ADSP-21160M | 4 Mbit | New! Features Single Instruction Multiple Data (SIMD) core architecture; optimized for multiprocessing with link ports, 64 bit external bus, and 14 channels of DMA |
| ADSP-21060 | 4 Mbit | Power house of the family; most memory; link ports for high speed data transfer and multi-processing |
| ADSP-21062 | 2 Mbit | Same features as the ADSP-21060, but with less internal memory (SRAM), for lower cost |
| ADSP-21061 | 1 Mbit | Low cost version used in the EZ-KIT Lite; less memory & no link ports; additional features in DMA for the serial port |
| ADSP-21065L | 544 kbit | A recent addition to the family; fast and very low cost ($10). Will attract many fixed point applications to the SHARC family |
| ADSP-21020 | -0- | Oldest member of the family. Contains the core processor, but no on-chip memory or I/O interface. Not quite a SHARC DSP. |

TABLE 29-1
Members of the SHARC family.

computers within a single chip. All they require to operate is a source of power, and some way to load a program into memory, such as an external PROM or data link.

Notice in Table 29-1 that even the low-end products have a very significant amount of memory. For instance, the ADSP-21065L has 544 kbits of internal SRAM. This is enough to hold 6-8 seconds of digitized speech (8k samples per second, 8 bits per sample). On the high-end of the family, the ADSP-21060 has a 4 Mbit memory. This is more than enough to store an entire digitized image (512×512 pixels, 8 bits per pixel). If you require even more memory, you easily add external SRAM (or slower memory) to any of these devices.

In addition to memory, there are also differences between these family members in their I/O sections. The ADSP-21060 and ADSP-21062 (the high-end) each have six *link ports*. These are 4 bit wide parallel connections for combining DSPs in multiprocessing systems, and other applications that require flexible high-speed I/O. The ADSP-21061 and ADSP-21065L (the low-end) do not have link ports, but feature more DMA channels to assist in their *serial port* operation. You will also see these part numbers with an "L" or "M" after them, such as "ADSP-21060L." This indicates that the device operates from a voltage lower than the traditional 5.0 volts. For